



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VIJAYALAKSHMI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

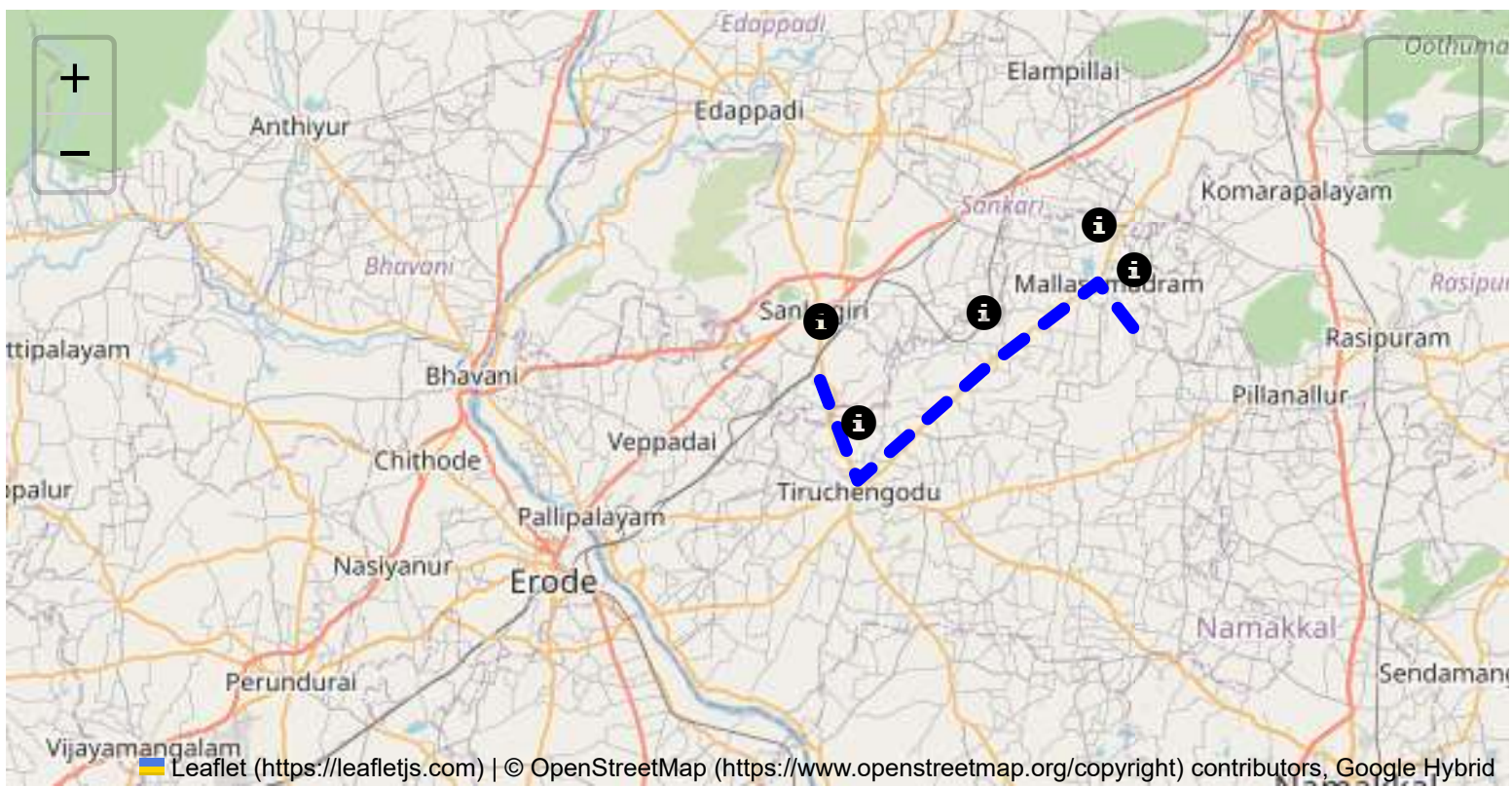
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

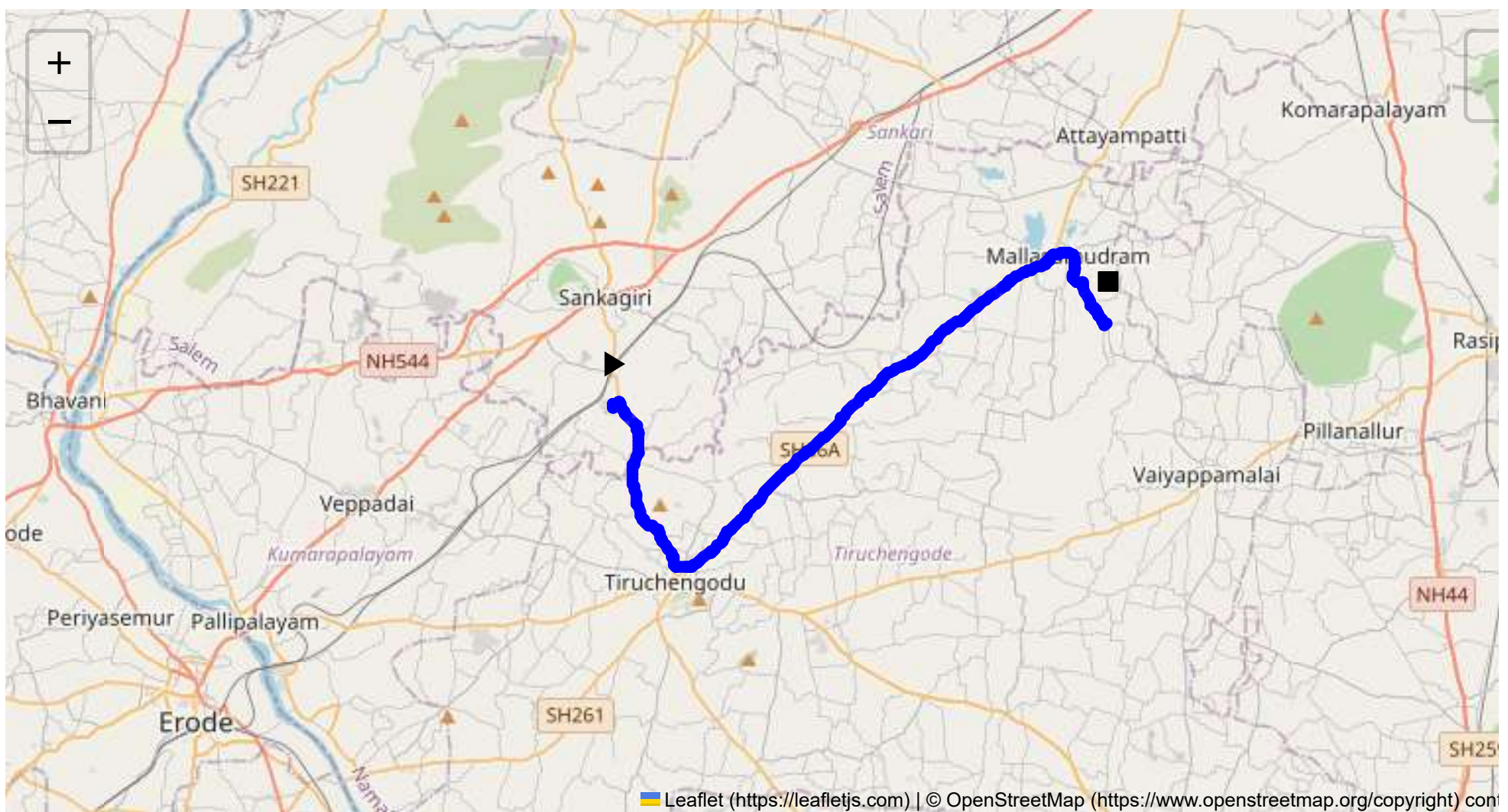
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 30.63 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.0 hours
Start: (11.4381, 77.8734)
End: (11.466244, 78.045265)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map:

- The route begins at CVQF+23W, Sangagiri, and proceeds southeast toward F28V+8RF, tk, Tiruchengodu. Key waypoints include Puthur, Tiruchengodu Central Bus Stand, and Mallasamudram, traversing through various local roads and state highways.

- Major roads: Tiruchengode - Salem Main Rd, crucial for connectivity between towns.

2. **Typical Weather Conditions and Hazards:**

- This region experiences a tropical climate with significant rainfall during the monsoon season, around June to September. Potential hazards include waterlogging and reduced visibility.
- Dry and hot weather from March to May might lead to heatwaves and road surface deterioration.

3. **Traffic Patterns:**

- Peak hours are typically from 8:00 to 10:00 AM and 5:00 to 7:00 PM due to local commuter and school traffic.
- Congestion-prone areas include the Tiruchengode Central Bus Stand and intersections along the Salem Main Rd.

4. **Road Quality and Infrastructure:**

- Roads are generally paved but may present quality issues like potholes, especially during the monsoon.
- Watch for narrow stretches near village entrances and certain congested Market areas in Tiruchengode.

5. **Alternative Routes for Emergencies:**

- In case of roadblocks, consider using local offshoots like branching from Tiruchengode - Erode Main Road and connecting back via the Namakkal road, although these may extend travel time.

6. **Local Regulations Affecting Hazardous Material Transport:**

- Enforcement of restrictions on the transport of hazardous materials through dense urban areas, active especially during daytime hours.
- Mandatory permits for the transport of hazardous materials, adherence to specific designated times.

7. **Historical Incidents Involving Heavy Vehicles:**

- Occasional reports of heavy vehicle accidents along the Tiruchengode - Salem Main Rd, usually attributed to speeding and road surface issues.
- No significant incidents involving hazardous materials in recent records but stay updated with local traffic advisories.

8. **Environmental Considerations and Sensitive Areas:**

- The area includes agricultural land; hence, be cautious of potential spills that can impact soil and crops.
- Proximity to residential areas necessitates caution in hazardous material handling to prevent air and water pollution.

9. **Communication Coverage:**

- Overall good coverage of cellular and data networks, with potential temporary dead zones in rural patches, mainly near Puthur.
- Advise carrying alternative communication devices like satellite phones for emergencies.

10. **Estimated Emergency Response Times:**

- Proximity to major towns like Tiruchengode enhances emergency response capabilities; average response time is around 30-45 minutes, but can vary based on the exact location and time of day.

11. Overall Summary of Risk Assessment:

- The route, while relatively short, traverses through areas with moderate traffic and occasional road quality issues. The key risks include congestion, especially near Tiruchengode, weather-related challenges during monsoon, and potential communications gaps. Strong emphasis on adherence to safety regulations and alternative route planning is essential for carrying hazardous materials. Regular communication with local authorities and updated GPS systems will mitigate many of the route's inherent risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38346, 77.89975	30 KM/Hr
6	Blind Spot	Blind Spot	11.44387, 77.96283	10 KM/Hr
7	Blind Spot	Blind Spot	11.44391, 77.96278	10 KM/Hr
8	Turn	High	11.44387, 77.96283	15 KM/Hr
9	Turn	High	11.49063, 78.02680	15 KM/Hr
10	Turn	Medium	11.46721, 78.04520	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
7	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
8	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38346, 77.89975



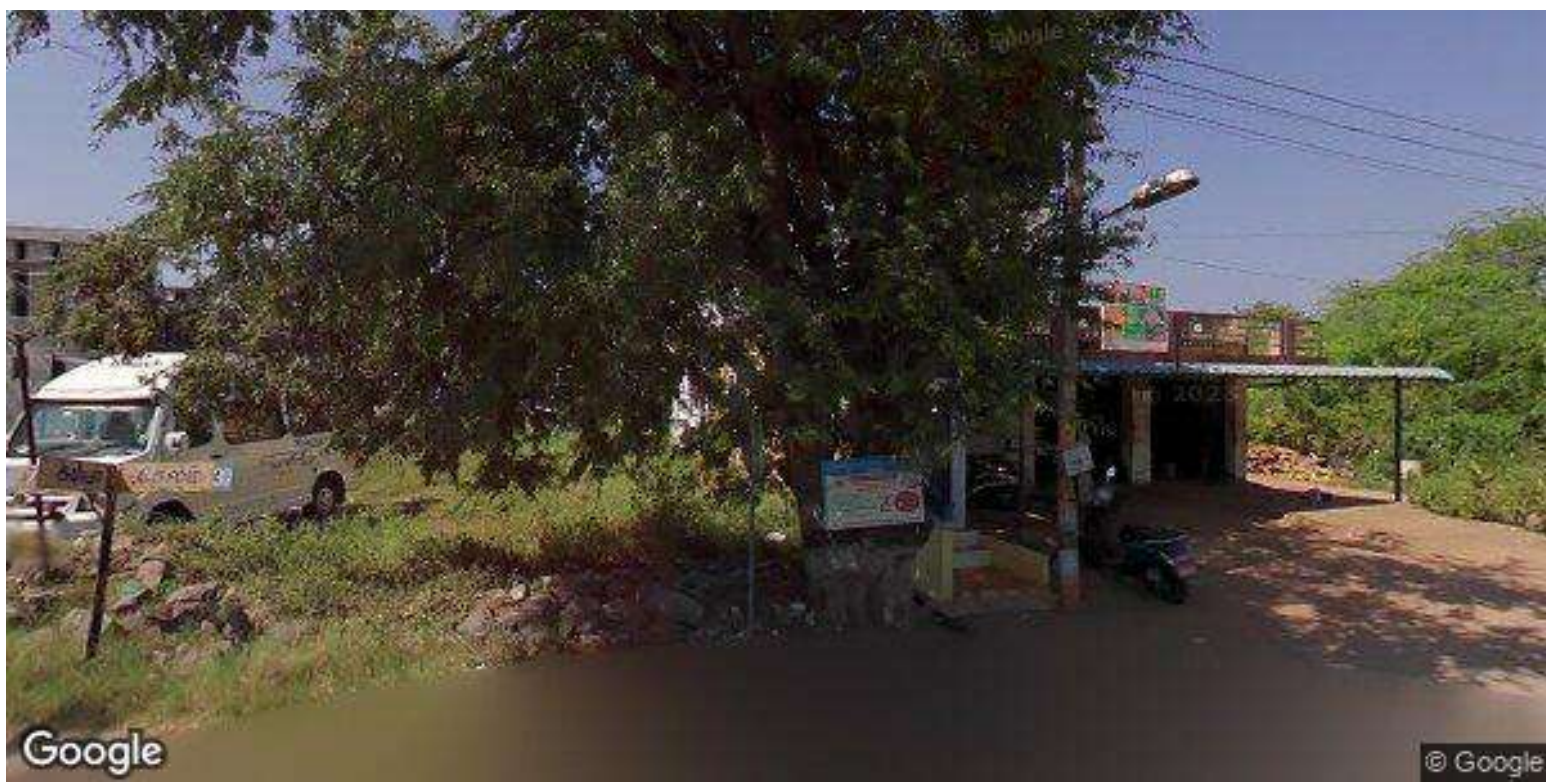
Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44387, 77.96283



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44391, 77.96278



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44387, 77.96283



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49063, 78.02680



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46721, 78.04520
